Index

Page numbers point to key discussions, chapter glossary entries, and reference tables. Select a page number to open the corresponding chapter or appendix PDF.

A

absolute zero, 250, 298
acceleration, 29, 54
acceleration due to gravity, 44, 54
AC current, 389
activity, 552, 592
AC voltage, 389
adiabatic process, 275, 298
air resistance, 51, 54
alpha, 535
alpha decay, 544, 592
alpha ray, 592
alternating current, 386, 389
AM. *See* amplitude modulation (AM)
ampere, 332, 352
Ampere's law, 376, 377, 389
amplitude, 156, 181
amplitude modulation (AM), 410, 450
Anger camera, 557, 592
angular acceleration, 190, 191, 209
angular momentum, 202, 209
angular velocity, 190, 209
antielectron, 548, 592
antimatter, 547, 592
antinode, 168, 181
apparent weight, 77, 97
Archimedes' principle, 227, 240
atomic mass, 540, 592
Atomic Masses, 607, 608, 609, 610, 611, 612
atomic number, 540, 592
audible range, 171, 181
average acceleration, 29, 54, 55
average speed, 26, 54
average velocity, 25, 54
axiom, 496, 524
axis of a polarizing filter, 444, 450

B

bandwidth, 410, 450
banked curve, 95, 97
baryon number, 585, 592
baryons, 587, 592
base units, 10
beat frequency, 169, 181
becquerel, 552, 592
Bernoulli's equation, 236, 240
Bernoulli's principle, 238, 240
beta, 535
beta decay, 546, 592
beta ray, 592
B-field, 369, 389
Biological Effects of Ionizing Radiation, 559
Biot-Savart law, 377, 389
birefringent, 448, 450
blackbody, 463, 486
blackbody radiation, 463, 486
Bohr radius, 482, 486
Bohr's Theory of the Hydrogen Atom, 480
Boltzmann constant, 254, 298
boson, 583, 592
bow wake, 179, 181
break-even, 592
breeder reactors, 592
breeding, 574, 592
bremsstrahlung, 417, 450
buoyant force, 227, 240

C

capacitor, 326, 352
carbon-14 dating, 551, 592
Carnot cycle, 282, 298
Carnot efficiency, 282, 298
Carnot engine, 282, 298
carrier wave, 410, 450
cathode-ray tube, 475, 486
Celsius, 250
Celsius scale, 298
center of mass, 90, 97
centripetal acceleration, 53, 54
centripetal force, 94, 97
change in entropy, 290, 298

characteristic X-ray, 417, 450
chart of the nuclides, 543, 592
chemical energy, 120, 128
classical physics, 6, 15
classical velocity addition, 524
closed system, 298
coefficient of performance, 289, 298
comets, 405
Common metric prefixes, 618
conduction, 258, 298
conductor, 326, 352
Conservation of energy, 105
conservation of mechanical energy, 118, 128
Conservation of Momentum, 139
conservation of momentum principle, 139, 148
conservation of total baryon number, 585, 592
conservation of total electron family number, 592
conservation of total muon family number, 592
conservative force, 113, 118, 128
constructive interference, 166, 181
convection, 258, 298
converging lens, 428, 450
converging mirror, 450
conversion factor, 11, 15
correspondence principle, 462, 486
Coulomb force, 316, 317, 352
Coulomb's law, 315, 352
criticality, 574, 592
critical mass, 573, 592
curie, 553, 592
Curie temperature, 366, 389
Current, 332
cyclical process, 278, 298

D

damping force, 160, 181
daughter, 544, 592
de Broglie wavelength, 469, 486
decay, 534, 544, 592
decay constant, 551, 592
decay equation, 545
decay series, 544, 592
deformation, 81, 97
degree Celsius, 250, 298
degree Fahrenheit, 250, 298
density, 219, 240
derived units, 15
destructive interference, 166, 181
direct current, 386, 389
direction of magnetic field lines, 369, 389
direction of polarization, 444, 450
Discovery of the Atomic Nucleus, 475
dispersion, 425, 450
displacement, 20, 54
distance, 21, 54
distance traveled, 21, 54
diverging lens, 429, 450
diverging mirror, 450
domains, 366, 389
Doppler Effect and Sonic Booms, 177
Doppler shift, 177, 181
dynamics, 61, 63, 97, 98

E

effective dose, 593
Einstein's Postulates, 496
elapsed time, 25, 54
elastic collision, 142, 148
Elastic Collisions in One Dimension, 142
elastic potential energy, 122, 128
electrical energy, 120, 128
electric charge, 311, 352
electric current, 332, 352
electric field, 317, 319, 352
electric field lines, 319, 352
electric field strength, 317, 318, 352
Electric Hazards and the Human Body, 348
Electricity, 309
electric polarization, 326, 352
electric potential, 322, 352
Electric Potential and Potential Energy, 321
electric power, 337, 352
Electric Power and Energy, 337
electromagnet, 367, 389
electromagnetic force, 310, 352
electromagnetic induction, 379, 389
electromagnetic spectrum, 407, 450
electromagnetic spectrum, applications, 409
electromagnetic waves, 403, 450
electromagnetism, 401, 450

electron, 312, 352
electron antineutrino, 547, 593
electron capture, 549, 593
electron family number, 585, 593
electron neutrino, 548, 593
electron volt, 352
electrostatic equilibrium, 326, 352
electrostatic force, 315, 352
electrostatic precipitators, 331, 352
electrostatics, 328, 352
electroweak theory, 588, 593
energy, 105, 106, 128
energy-level diagram, 482, 486
English units, 9, 15
entropy, 290, 298
entropy, statistical interpretation, 294
equivalent dose, 593
external force, 66, 97
extremely low frequency (ELF), 410, 450

F

Fahrenheit, 250
Fahrenheit scale, 298
Falling Objects, 44
Faraday cage, 328, 352
Faraday's law of induction, 381, 389
fermion, 583, 593
ferromagnetic, 366, 389
Ferromagnets and Electromagnets, 366
Feynman diagram, 580, 593
field, 352
first law of thermodynamics, 271, 298
first postulate of special relativity, 497, 524
Fission, 571
fission fragments, 572, 593
flow rate, 233, 240
fluids, 217, 218, 240
FM. *See* frequency modulation (FM)
focal length, 428, 450
focal point, 428, 450
force, 63, 97, 98
Forced Oscillations and Resonance, 160
force field, 317
free-body diagram, 66, 97
free charge, 326, 352
free-fall, 44, 54
frequency, 154, 181
frequency modulation (FM), 410, 450
friction, 84, 97
fundamental forces, 580
fundamental frequency, 168, 181
fundamental units, 15
Fusion, 565

G

gamma, 535
gamma camera, 557, 593
gamma decay, 549, 593
gamma ray, 407, 450
gauge boson, 583, 593
gauss, 371, 389
Geiger tube, 538, 593
Glossary of Key Symbols and Notation, 623
gluons, 580, 589, 593
grand unified theory, 588, 593
gravitational constant, 90, 97
gravitational potential energy, 113, 128
gray (Gy), 560, 593
Greek alphabet, 618
grounded, 329, 352
ground state, 482
GUT. *See* grand unified theory
GUTs: The Unification of Forces, 588
gyroscopic effects, 207

H

hadrons, 583, 593
half-life, 550, 593
Half-Life and Activity, 550
hearing, 171, 181
heat, 256, 298
heat engine, 273, 298
heat engine processes, 273
heat of sublimation, 269, 298
heat pump, 298
heat pumps and refrigerators, 286
heat pump's coefficient of performance, 287
heat transfer, 247
Heat Transfer Methods, 258
Heisenberg's uncertainty principle, 473, 486

hertz, 154, 181
Higgs boson, 591, 593
high dose, 561, 593
Hooke's law. *See* Spring Force: Hooke's Law
horizontally polarized, 444, 450
hormesis, 562, 593
horsepower, 125, 128
human metabolism, 298
hydrogen-like ion, 486

I

ideal banking, 96, 97
ideal gas law, 253, 254, 298
ignition, 593
Image distance, 432
Image Formation by Lenses, 428
Image Formation by Mirrors, 438
Important Constants, 617
impulse, 137, 148
Impulse and Momentum, 133
index of refraction, 423, 450
induced current, 379, 389
Induced Voltage and Magnetic Flux, 379
inelastic collision, 145, 148
Inelastic Collisions in One Dimension, 145
inertia, 64, 97, 98
inertial confinement, 568, 593
inertial frame of reference, 497, 524
infrared radiation (IR), 413, 450
infrasound, 172, 181
ink jet printer, 330
instantaneous acceleration, 30, 54
instantaneous speed, 26, 54
instantaneous velocity, 26, 54
internal energy, 272, 298
ionizing radiation, 536, 593
IR. *See* infrared radiation (IR)
irreversible process, 277, 298
isobaric process, 273, 274, 298
isochoric process, 274, 298
isolated system, 298
isothermal process, 275, 298
isotopes, 541, 593

J

joule, 107, 128
Joule's law, 342, 352

K

Kelvin, 250
Kelvin scale, 299
kilocalorie, 256, 299
kilogram, 10, 15
kilowatt-hour, 126, 128
kinematics, 19, 54
kinetic energy, 110, 128
kinetic friction, 84, 85, 97

L

Laser printers, 330
latent heat coefficient, 265, 299
law, 5, 15
law of conservation of angular momentum, 204, 209
law of conservation of charge, 311, 314, 352
law of conservation of energy, 120, 128
law of inertia, 64, 97, 98
law of reflection, 450
length contraction, 506, 507, 524
lens power, 429, 450
Lenz's law, 389
leptons, 583, 593
lever arm, 194, 209
Light, 399
linear hypothesis, 562, 593
linear momentum, 134, 148
Linear Momentum and Force, 134
liquid drop model, 572, 593
liter, 233, 240
longitudinal wave, 165, 181
low dose, 561, 593
luminiferous aether, 497, 524

M

macrostate, 294, 299
magic numbers, 543, 593
magnetic confinement, 568, 594
magnetic field, 369, 389
magnetic field lines, 369, 389

Magnetic Fields and Magnetic Field Lines, 369
magnetic field strength, 371, 389
magnetic flux, 379, 389
magnetic force, 389
Magnetic Force on a Current-Carrying Conductor, 373
magnetic monopoles, 368, 389
Magnetism, 363
magnetized, 366, 389
Magnets, 364
magnification, 433, 450
mass, 64, 97, 98
mass-energy equivalence. *See* Relativistic Energy
mass number, 540, 594
matter, wave nature, 469
Maxwell's equations, 400, 451
mechanical energy, 119, 128
mechanical equivalent of heat, 256, 299
Medical Imaging and Diagnostics, 556
meson, 587, 594
meter, 10, 15
metric system, 11, 15
Michelson-Morley experiment, 498, 524
microgravity, 92, 97
microshock sensitive, 351, 352
microstate, 294, 299
microwaves, 412, 451
mirror, 451
model, 4, 15
moderate dose, 561, 594
modern physics, 7, 15
moment of inertia, 195, 209
motion, 51, 54
Motion Equations for Constant Acceleration in One Dimension, 36
motor, 375, 389
Motors and Meters, 375
moving-coil meter, 375, 389
muon family number, 585, 594

N

natural frequency, 160, 181
net external force, 66, 97
net work, 109, 128
neutrino, 546, 594
neutron, 540, 594
neutron activation, 567, 594
neutron-induced fission, 572, 594
newton, 68
newton-meters, 107
Newton's first law of motion, 64, 97, 98
Newton's second law of motion, 66, 97
Newton's third law of motion, 73, 97, 98
Newton's universal law of gravitation, 89, 97
nodes, 168, 181
normal force, 77, 97
Normal Force and Tension, 77
north magnetic pole, 364, 389
Nuclear and Particle Physics, 533
Nuclear Decay and Conservation Laws, 544
nuclear energy, 120, 128
nuclear fission, 571, 594
nuclear fusion, 565, 594
nuclear radiation, 534, 594
Nuclear Radioactivity, 534
nuclear reaction energy, 545, 594
Nuclear Weapons, 576
nucleons, 540, 594
nucleus, 594
nuclide, 540, 594

O

ohm, 335, 352
ohmic, 335, 352
Ohm's law, 335, 352
optically active, 447, 451
order of magnitude, 11, 15
oscillate, 153, 181
Oscillations and Waves, 153
Other Units, 620
Otto cycle, 280, 299
overtones, 168, 181

P

parallel, 342, 353
parallel-plate capacitor, 326, 353
parent, 544, 594
Particles, Patterns, and Conservation Laws, 582
perfectly inelastic collision, 145, 148
period, 154, 181

Period and Frequency in Oscillations, 154
periodic motion, 154, 181
permeability of free space, 376, 389
PET. *See* positron emission tomography (PET)
Phase Change and Latent Heat, 265
photoconductor, 329, 353
photoelectric effect, 465, 486
photomultiplier, 539
photomultiplier tube, 594
photon, 466, 486
photon energy, 466, 486
Physical Quantities and Units, 9
physical quantity, 9, 15
physics, 3, 15
Physics: An Introduction, 1, 2
pitch, 173, 181
Planck Law, 463, 486
Planck's constant, 463, 486
planetary model of the atom, 479, 486
point charge, 317, 353
polarization, 444, 451
polarized light, 444, 451
position, 20, 54
positron, 548, 594
positron decay, 548, 594
positron emission tomography (PET), 558, 594
postulate, 496, 524
potential difference, 322, 353
potential energy, 118, 128
power, 124, 128
precession, 208, 209
pressure, 222, 224, 240
pressure due to the weight of fluid, 224, 240
principle of relativity, 497, 524
probability distribution, 486
probability wave, 472, 486
Production and Properties of Electromagnetic Waves, 403
projectile, 51, 54
projectile motion, 51, 54
proper length, 506, 524
proper time, 501, 524
proton, 312, 353
proton-proton cycle, 566, 594

Q

QCD. *See* quantum chromodynamics
QED. *See* quantum electrodynamics
quality factor, 560, 594
quantization, wave explanation, 484
quantized, 462, 486
quantum chromodynamics, 590, 594
quantum electrodynamics, 580, 594
quantum mechanics, 461, 462, 486

R

rad, 559, 594
radar, 412, 451
radian, 190, 209
radiant energy, 120, 128
radiation, 258, 299
Radiation Detection and Detectors, 538
radiation detector, 538, 594
radiation pressure, 405, 451
radiation weighting factor, 594
radioactive, 534, 594
radioactive dating, 551, 594
radioactivity, 534, 594
radiopharmaceutical, 556, 594
radio waves, 409, 451
rainbow, 451
range, 51, 54
range of radiation, 536, 595
Ray tracing, 430
RBE. *See* relative biological effectiveness (RBE)
real image, 432, 451
Reflection, 420
refraction, 422, 451
relative biological effectiveness (RBE), 560, 595
Relativistic Addition of Velocities, 510
relativistic Doppler effects, 524, 526
Relativistic Energy, 517
relativistic kinetic energy, 520, 524
relativistic momentum, 515, 524
relativistic velocity addition, 511, 524
relativity, 496, 524
resistance, 335, 353
resistor, 340, 353
Resistors in Series and Parallel, 340

resonance, 160, 181
rest energy, 517, 524
rest mass, 515, 524
restoring force, 81, 97
reversible process, 275, 299
right-hand rule, 207, 209
right hand rule 1 (RHR-1), 372, 390
right hand rule 2 (RHR-2), 376, 390
rms, 387, 390
roentgen equivalent man (rem), 560, 595
Rotation, 189
rotational inertia, 194, 195, 209
rotational kinetic energy, 197, 209
Rydberg constant, 480, 486

S

scalar, 23, 54
scientific method, 7, 15
scintillators, 538, 595
second, 10, 15
second law of thermodynamics, 277, 278, 299
second law of thermodynamics stated in terms of entropy, 292, 299
second postulate of special relativity, 498, 524
Selected British Units, 619
Selected Radioactive Isotopes, 613, 614, 615
series, 340, 353
shielding, 563, 595
shock hazard, 348, 353
short circuit, 348, 353
sidebands, 410, 451
sievert, 560, 595
simple harmonic motion, 156, 181
simple harmonic oscillator, 156, 181
Simultaneity And Time Dilation, 499
single-photon-emission computed tomography (SPECT), 557, 595
SI units, 9, 15, 619
Snell's law. *See* refraction
solar sail, 406, 451
Solar System Data, 618
solenoid, 377, 390
solid-state radiation detectors, 539, 595
sonic boom, 179, 181
sound, 171, 181
south magnetic pole, 364, 390
space-time diagram, 500, 524
special relativity, 495, 497, 525
specific gravity, 230, 240
specific heat, 260, 299
SPECT. *See* single-photon-emission computed tomography (SPECT)
speed of light, 451
Speed of Sound, Frequency, and Wavelength, 173
spring constant, 81, 97
Spring Force: Hooke's Law, 81
Spring Potential Energy, 122
Standard Model, 590, 595
standing wave, 167, 181
static electricity, 309, 353
static friction, 84, 85, 98
step-down transformer, 384, 390
step-up transformer, 384, 390
strangeness, 586, 595
sublimation, 269, 299
Submicroscopic Masses, 618
Substructure of the Nucleus, 540
supercriticality, 574, 595
superposition, 166, 181
superstring theory, 591, 595
symbols and notation. *See* Glossary of Key Symbols and Notation
system, 66, 98

T

tagged, 556, 595
tangential acceleration, 192, 209
tau family number, 595
Television, 411
temperature, 248, 299
Temperature Change and Heat Capacity, 259
tension, 77, 98
tesla, 371, 390
test charge, 317, 353
theory, 4, 15
thermal energy, 120, 128
thermal equilibrium, 252, 299
thermal hazard, 348, 353
Thermal Physics, 247
thermodynamic process, 274

thin lens, 430
thin lens equations, 432
thrust, 73, 98
time, 25, 54
time dilation, 500, 525
Time, Velocity, and Speed, 25
torque, 194, 209
total energy, 517, 525
trajectory, 51, 54
transformer, 383, 390
transformer equation, 384, 390
transverse wave, 164, 181
twin paradox, 525

U

UHF. *See* ultra-high frequency (UHF)
ultracentrifuge, 54
ultra-high frequency (UHF), 411, 451
ultrasound, 171, 181
ultraviolet catastrophe, 463, 486
ultraviolet radiation (UV), 415, 451
uncertainty in momentum, 486
uncertainty in position, 486
Uncertainty Principle, 472
uniform circular motion, 53, 54
units, 9, 15
unpolarized, 444, 451
Useful Formulae, 621
Useful Information, 617
UV. *See* ultraviolet radiation (UV)

V

Van de Graaff generator, 328, 353
vector, 23, 54, 319
Vectors, Scalars, and Coordinate Systems, 23
vertically polarized, 444, 451
very high frequency (VHF), 411, 451
VHF. *See* very high frequency (VHF)
virtual image, 434, 451
visible light, 414, 451
voltage, 322
voltage drop, 340, 353

W

watt, 124, 128
wave, 153, 163, 167, 181
Wave Interference: Standing Waves and Beats, 166
wavelength, 163, 182
wave velocity, 163, 182
weight, 68, 77, 98
work, 106, 128
Work and Energy, 105
work-energy theorem, 109, 110, 128
work function, 467, 486
worldline, 500, 525

X

xerography, 329, 353
X-ray, 417, 451

Z

zeroth law of thermodynamics, 252, 299